Methodology

3.1 Conceptual Architecture of Hybrid Transformer-CNN

The proposed framework is designed as a UAV-based semantic segmentation system for detecting potato leaf disease symptoms from RGB drone imagery. The main objective of the architecture is to generate a pixel-wise field health map from high-resolution UAV photographs captured at different heights. Instead of classifying an entire image as healthy or diseased, the framework identifies plant and disease regions at the pixel level, allowing disease density and hotspot severity to be calculated from the predicted mask. To achieve reliable detection across all four UAV capture heights, four independent height-specific models are trained — one each for 2 m, 7 m, 10 m, and 12 m. At inference time the appropriate model is selected automatically based on the altitude metadata embedded in the drone image.

The primary model architecture follows a Hybrid Transformer-CNN design based on a SegFormer MiT-B3 encoder and a U-Net decoder. The SegFormer MiT-B3 encoder extracts high-level contextual features from UAV image tiles, which is important because potato disease symptoms may appear under different heights, lighting conditions, canopy densities, and background patterns. The U-Net decoder reconstructs the spatial structure of the prediction and helps recover disease boundaries at the pixel level. The final segmentation output is planned for four classes: Background, Healthy, Early Blight, and Late Blight. Background represents soil, shadows, non-target objects, and other non-plant regions; Healthy represents visible healthy potato foliage; Early Blight represents disease regions with characteristic dark necrotic lesions and concentric spotting; and Late Blight represents rapidly spreading water-soaked or necrotic disease regions. This four-class structure allows the framework to separate healthy plant tissue from major foliar disease symptoms while still excluding non-plant background pixels.

In the planned workflow, each UAV image will be first organised by capture height using the DJI EXIF RelativeAltitude metadata. A separate height-specific dataset is then built for each altitude (2 m, 7 m, 10 m, and 12 m), and a dedicated Hybrid MiT-B3 + U-Net model is trained independently on each subset. During inference, the input image altitude is read from the embedded metadata, and the corresponding height-specific model is loaded to produce the four-class segmentation output. Softmax converts the raw logits into class probabilities, and Argmax selects the most probable class for each pixel. The complete architecture therefore connects UAV image acquisition, altitude-based data partitioning, per-height hybrid deep learning training, tiled inference, and field-level disease severity reporting into one reproducible workflow.

Note: the implementation must be configured for four output classes before final training so that the code, masks, and methodology remain consistent.

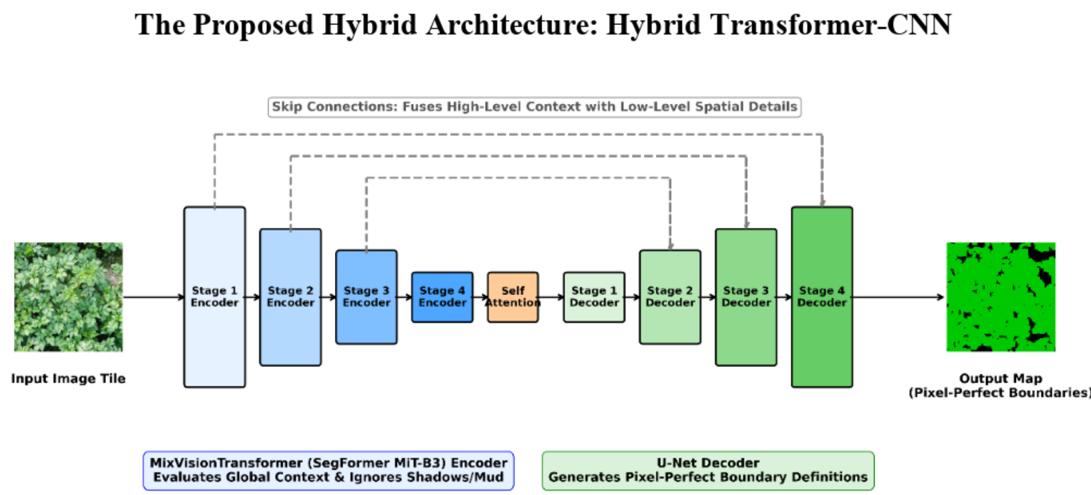

Fig 3.1: Hybrid Model Architecture (Hybrid Transformer-CNN)

3.2 Study Area

The study was conducted in potato-growing fields located around Hajee Mohammad Danesh Science and Technology University (HSTU), Dinajpur, Bangladesh. Image collection was carried out from several nearby agricultural areas, including Nashipur, Basherhat, Gopalganj, and surrounding farmer-managed fields of Dinajpur. These locations were selected because they represent real potato production environments where crop health, field size, irrigation condition, and disease occurrence vary from plot to plot.

Field image acquisition was conducted from 12 November 2025 to 20 January 2026, covering the potato-growing period from germination toward production. Although the collection period falls within the Rabi potato season in Bangladesh, most image-capture days were sunny, which provided clear visibility for UAV-based RGB image acquisition. Images were collected throughout the crop season, but visible disease symptoms were mainly observed when the plants were approximately 14 to 30 days old.

Consistent collection of diseased plant images was difficult under field conditions. Farmers in the selected areas commonly applied pesticides and fungicides, which reduced visible disease incidence in several plots. In addition, plants suspected to be affected by PLRV were often removed from the field soon after observation. For this reason, the dataset was collected from multiple fields and multiple potato varieties to increase field-level diversity and improve the practical reliability of the model.

Fig 3.2: UAV image collection from a farmer's field

The following satellite views present the approximate study-field locations and mapped boundaries used to support the description of the study area. The overview map shows the distribution of selected fields, while the closer views show individual field boundaries where UAV image acquisition was performed.

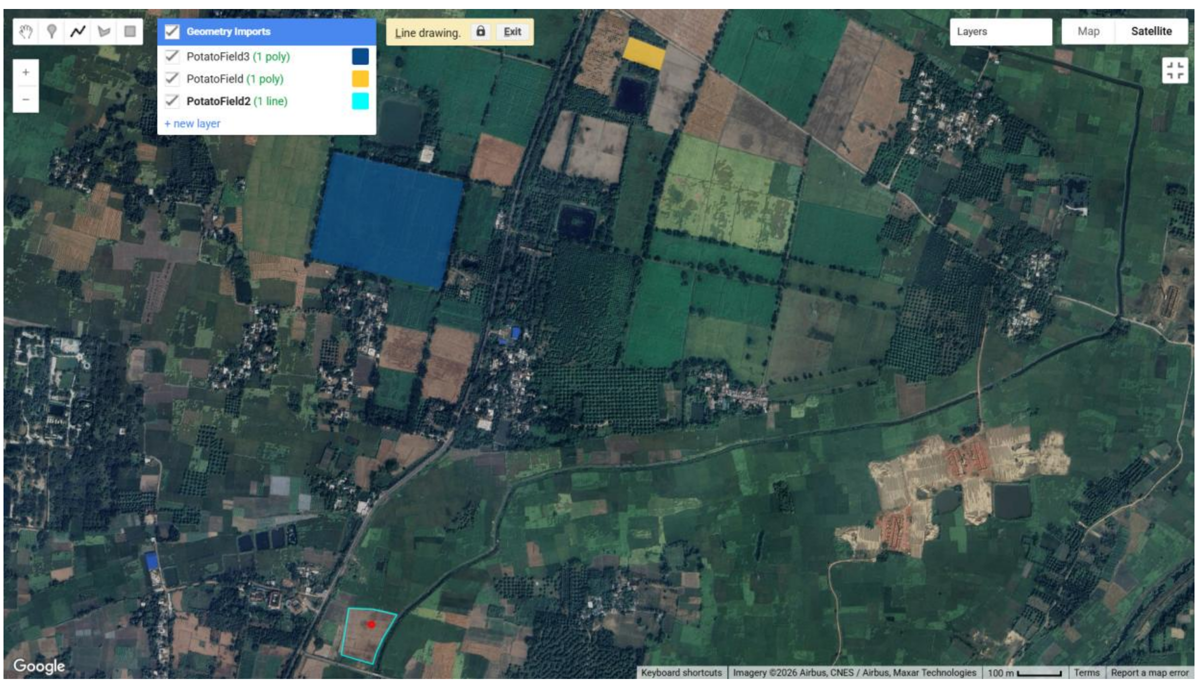

Fig 3.3 : Satellite overview of selected potato field locations around the HSTU study area.

Fig 3.4 : Satellite view of a selected potato field boundary used for UAV image acquisition.

Fig 3.5 : Close satellite view of an additional selected potato field boundary used during field sampling.

The selected fields were generally characterized by flat agricultural terrain, farmer-managed irrigation, and common local agronomic practices. Satellite views were added to document the spatial context of the study fields and to show the approximate field boundaries used during UAV image acquisition. These maps also help clarify that the dataset was not collected from a single controlled experimental plot; rather, it reflects several real production fields around the university area.

The study included several potato varieties commonly found in the surveyed Dinajpur fields, namely BARI Alu-7, BARI Alu-25 (Asterix), Cardinal, Diamant, and local varieties locally referred to as Deshi. Including multiple varieties was necessary because disease symptoms and canopy appearance may vary across cultivars, and a UAV-based model should remain reliable under such practical field variation.

Table 3.1: Five Potato Varieties That are Currently in The Dinajpur Sadar Upazila

Potato Variety	Characteristics and Relevance
BARI Alu-7	BARI-released high-yielding potato variety observed in the study-area fields.
BARI Alu-25 (Asterix)	Commercially cultivated variety, locally known as Asterix, included to capture varietal differences in canopy and disease appearance.
Cardinal	Commonly grown variety in Dinajpur and included because of its field availability during data collection.
Diamant	Widely cultivated variety included to increase dataset diversity across farmer-managed plots.
Local varieties (Deshi)	Farmer-maintained local varieties included to represent non-uniform field conditions and locally adapted crop types.

3.3 UAV Platform and Camera Specifications

Aerial imagery of the potato fields was captured using a DJI Mini 2 quadcopter equipped with an RGB camera. The images used in this study were saved as JPEG photographs with a resolution of 4000 x 2250 pixels. The DJI Mini 2 was selected because its lightweight body, stabilized camera, and rapid field deployment capability made it suitable for collecting low-altitude images across several farmer-managed plots around HSTU and Dinajpur.

During image acquisition, the UAV was manually positioned above the selected potato plants and images were captured vertically from four heights: 2 m, 7 m, 10 m, and 12 m. For each selected field position, three photographs were captured at every height. This means the same or closely similar crop area was photographed repeatedly from multiple heights, allowing the dataset to represent disease symptoms under different image scales while keeping the field context as consistent as possible.

This manual multi-altitude capture strategy was used because the study aimed to improve model reliability across different UAV image-capture heights while keeping the field location as consistent as possible. Capturing the same or similar crop area from multiple altitudes supports later scale-robustness analysis, since the same disease symptoms may appear at different pixel sizes depending on camera height.

The onboard camera is stabilized by a 3-axis mechanical gimbal, which helps reduce vibration and sudden changes in camera attitude during low-altitude field photography. Images were captured as RGB JPEG files rather than multispectral or thermal imagery; therefore, the proposed disease detection pipeline is based on visible symptoms such as leaf color, lesion pattern, necrosis, chlorosis, and leaf rolling. This RGB-only setup keeps the system low-cost and practical for local field use, but it also means that visually similar disease and stress symptoms require careful annotation and laboratory support.

Fig 3.6 : DJI Mini 2

Fig 3.7 : DJI Mini 2 SE box contents — drone body, DJI RC-N1 remote controller, intelligent flight batteries, gimbal protector, spare propellers, charging hub, shoulder bag, and accessories.

Table 3.2: UAV Specification

Manufacturer / Model	DJI Mini 2
UAV type	Quadcopter (foldable)
Maximum take-off weight	249 g
Camera	1/2.3-inch CMOS, 12 MP
Image format	JPEG
Field of view (FOV)	83° (equivalent focal length: 24 mm)
Gimbal stabilization	3-axis mechanical gimbal
Maximum flight time	~31 minutes per battery charge
Positioning system	GPS + GLONASS dual-satellite
Maximum wind resistance	Level 5 (8.0–10.7 m/s)
Image resolution	4000 x 2250 pixels (RGB JPEG images used in this study)

Table 3.3: UAV Image Acquisition Parameters and Ground Sampling Distance (GSD) at Various Flight Altitudes

Altitude	Photos Captured per Selected Position	Image Resolution	Approx. GSD
2 m	3 photos	4000 x 2250 pixels	0.09 cm/pixel
7 m	3 photos	4000 x 2250 pixels	0.31 cm/pixel
10 m	3 photos	4000 x 2250 pixels	0.44 cm/pixel
12 m	3 photos	4000 x 2250 pixels	0.53 cm/pixel

3.4 Experimental Procedure

The experimental procedure was designed to connect real-field UAV image collection with a reproducible deep learning workflow. The complete procedure consisted of field selection, manual multi-height image acquisition, disease confirmation support, annotation, automated preprocessing, model training, and final evaluation. This structure ensures that the methodology describes the full research process rather than only the image capture stage.

Potato fields around HSTU and nearby Dinajpur locations were selected based on crop availability and visible disease symptoms. Because disease incidence was not consistent in every field due to pesticide and fungicide use, images were collected from several farmer-managed plots. Crop observation covered stages from germination toward production, with disease symptoms most commonly observed when plants were approximately 14 to 30 days old.

3.4.1 UAV Image Acquisition

A core methodological component of this study is the use of four low-altitude image-capture heights: 2 m, 7 m, 10 m, and 12 m above ground level. These heights were included to make the proposed model more reliable under different UAV operating conditions rather than limiting the dataset to a single fixed height. Because farmers and field operators may capture images from different heights depending on field size, battery condition, crop density, and field accessibility, the dataset intentionally includes multi-height imagery so that the model can learn potato disease symptoms at different image scales.

For each selected field position, the UAV was manually positioned above the crop canopy and three photographs were captured at every height. Images were taken vertically in a top-down orientation without intentional oblique angle. Capturing the same or closely similar crop area from 2 m, 7 m, 10 m, and 12 m created overlapping multi-altitude visual coverage while keeping the field condition as consistent as possible.

Approximate Ground Sampling Distance (GSD) values were estimated using the DJI Mini 2 field of view and flight height. These values should be treated as approximate because actual GSD can vary with camera angle, terrain height, and image cropping.

This manual multi-altitude acquisition strategy was designed to improve model robustness across practical field-capture conditions. By collecting images of the same or similar crop areas from multiple heights, the dataset can expose the model to variation in apparent leaf size, lesion visibility, canopy coverage, and background proportion. This is expected to help the trained model generalize better when future UAV images are captured from different heights.

Lower-altitude images, especially those captured at 2 m, provide clearer leaf-level detail, while higher-altitude images such as 10 m and 12 m include more surrounding canopy and field context. Including all four heights therefore supports reliability across both close-range disease observation and wider field scouting situations.

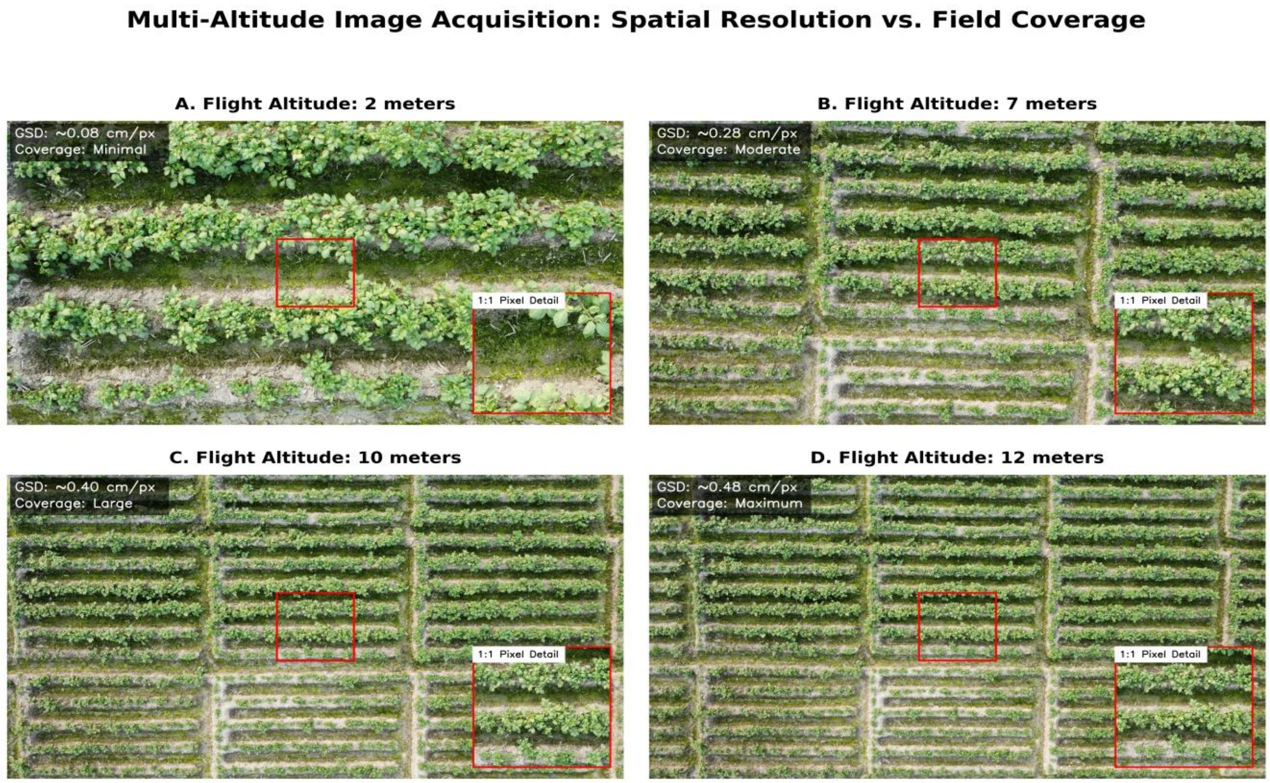

Fig 3.6: Multi-altitude UAV image acquisition for model reliability across different capture heights.

3.4.2 Dataset Preparation

After the initial collection of high-resolution UAV images, the dataset required rigorous labeling to prepare it for the semantic segmentation models. The raw images were imported into the Roboflow platform, which was utilized as the primary annotation environment due to its robust polygon-drawing tools and dataset management capabilities.

During the annotation phase, domain experts manually traced precise polygon boundaries around the regions of interest. The annotations were specifically categorized into three primary classes: Healthy, Early Blight, and Late Blight. Any unannotated pixels in the imagery were implicitly treated as the background. This polygon-based approach ensures that the intricate shapes of the potato leaves and disease lesions are captured with high fidelity.

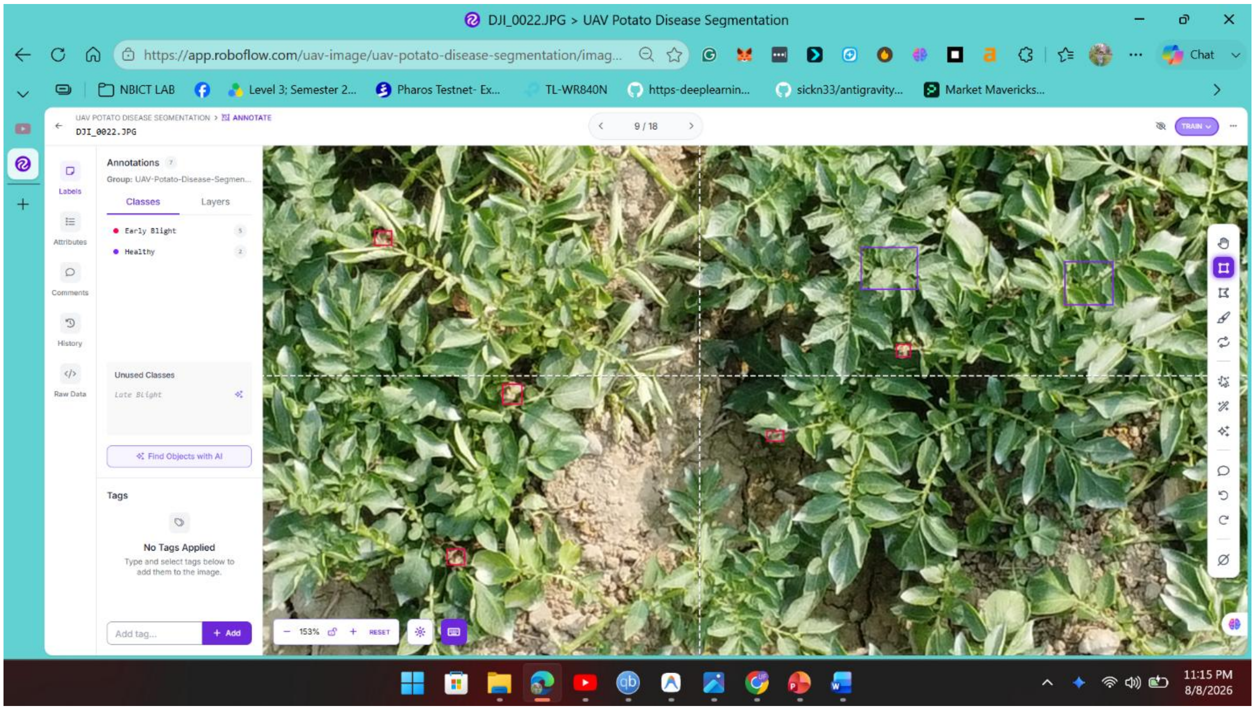

Fig 3.7: UAV Image Annotation In Roboflow

Once the manual labeling was complete, the dataset was exported from Roboflow in a semantic mask format (or COCO JSON) for local preprocessing. A custom Python-based preprocessing pipeline was developed to bridge the gap between the raw Roboflow export and the strict tensor requirements of the PyTorch-based neural network. The pipeline executed the following automated steps:The polygon coordinates (or semantic colors) exported from Roboflow were converted into strict class-index masks. The pipeline ensured that the unannotated regions were automatically assigned an index of 0 (Background), while Healthy, Early Blight, and Late Blight were mapped to indices 1, 2, and 3, respectively.To properly evaluate the model, the pipeline randomized the data and stratified it into three distinct subsets: 80% for training the model, 10% for validation during training epochs, and 10% reserved for final unseen testing.

Table 3.3: Dataset partitioning ratio.

Groups	**Percentage (%)**
Training	80%
Validation	10%
Testing	10%

Table 3.4: Dataset annotation categories.

Class	Description
Healthy	Visible healthy potato foliage without disease symptoms.
Early Blight	Disease-affected regions showing early blight symptoms such as dark necrotic spots and concentric lesion patterns.
Late Blight	Disease-affected regions showing late blight symptoms such as water-soaked, rapidly spreading, or necrotic leaf tissue.

3.4.3 Disease Confirmation and Laboratory Support

To improve the reliability of disease labeling, suspected disease samples were further examined with support from the Faculty of Agriculture at Hajee Mohammad Danesh Science and Technology University (HSTU). Laboratory confirmation was conducted by an experienced laboratory technician with the required practical skill to handle plant-disease diagnostic work. This step helped reduce uncertainty in visually similar disease symptoms and provided additional confidence before disease-affected samples were included in the dataset.

The laboratory examination was especially important for samples where field-level visual diagnosis was uncertain. Because symptoms such as leaf rolling, chlorosis, necrotic spots, and stress-related discoloration can overlap across different potato diseases and abiotic conditions, laboratory support from the Faculty of Agriculture was used as an additional verification step. The specific laboratory test name, equipment, and diagnostic protocol should be inserted here after confirmation from the technician or supervisor.

To ensure unbiased evaluation and prevent data leakage, the dataset will be partitioned independently for each of the four UAV capture heights. All tiles generated from the same original UAV image will remain within the same subset (training, validation, or test) to prevent spatial leakage between subsets. The planned partitioning ratio is 80% training, 10% validation, and 10% Based on approximately 14,000 total candidate tiles from all four heights combined, the per-height tile distribution is expected to vary because higher altitudes cover more field area per image while lower altitudes (particularly 2 m) produce images with fewer total tiles per session.

3.5 Model Architecture

Hybrid Transformer-CNN (SegFormer MiT-B3 + U-Net)

The multi-altitude drone dataset will be divided into four independent altitude-specific subsets before model training begins. Each subset contains tiles extracted from UAV images captured at one specific height (2 m, 7 m, 10 m, or 12 m). Within each altitude subset, the data will be further partitioned into training, validation, and test splits at an 80/10/10 ratio at the original-image level. Preprocessing for all four subsets uses the same spatial tiling engine to extract 640 x 640 pixel patches with 20% overlap. The training splits are augmented using brightness, contrast, 90-degree rotation, fog/noise simulation, and scale-related transformations. Validation data will be used only for model selection and hyperparameter monitoring, while the test split will be reserved strictly for final per-height performance reporting.

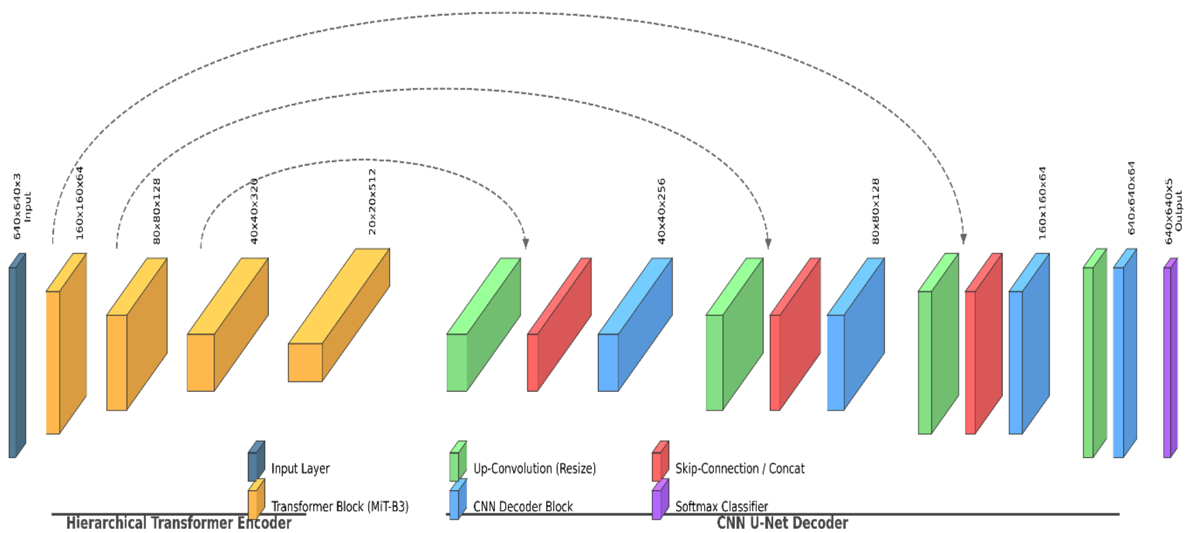

Fig 3.8: 3D Architectural Hybrid Transformer-CNN (SegFormer MiT-B3 + U-Net) model

A Hybrid Transformer-CNN architecture is employed as the core segmentation engine for each height-specific model. The MixVisionTransformer (SegFormer MiT-B3), pre-trained on ImageNet, is used as the shared encoder backbone across all four models. It captures global contextual features from 640 x 640 pixel UAV tiles, which is important because disease symptoms appear with different canopy coverage and texture depending on the capture height. A U-Net decoder reconstructs the pixel-level segmentation mask from these features by combining high-level semantic information with lower-level spatial detail, enabling more precise boundary recovery for irregular disease regions. The final output layer performs four-class pixel-wise segmentation across Background (class 0), Healthy (class 1), Early Blight (class 2), and Late Blight (class 3). While all four models share this identical architecture and ImageNet initialisation, each model is fine-tuned exclusively on its corresponding altitude-specific subset, so its learned weights reflect the visual characteristics of disease at that particular GSD level. Four independent weight files are therefore produced: best_hybrid_model_2m.pth, best_hybrid_model_7m.pth, best_hybrid_model_10m.pth, and best_hybrid_model_12m.pth.

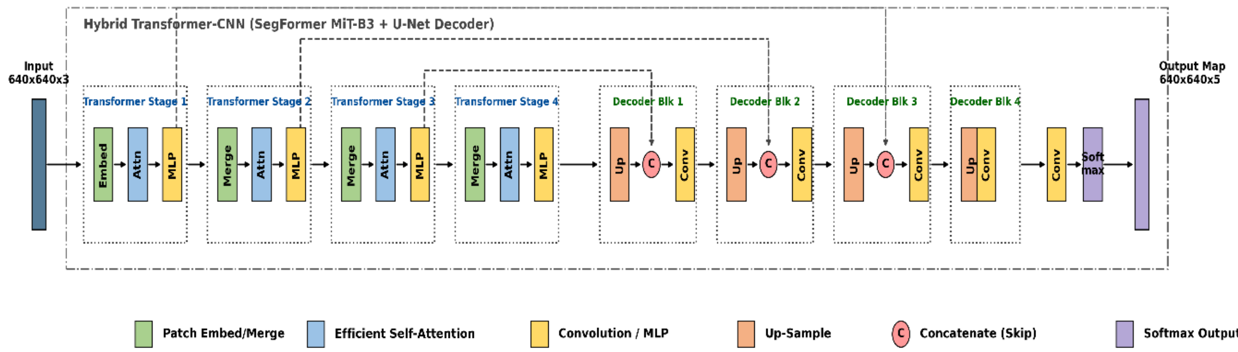

Fig 3.9: Internal operations of the Hybrid Transformer-CNN architecture.

Each of the four height-specific models will be configured with the Adam optimiser and an initial learning rate of 0.0001. The loss function will be selected before final training based on class imbalance within each altitude subset: categorical cross-entropy will serve as the baseline, while a combined Dice and Focal loss is recommended for the final experiment because disease pixels are expected to occupy a smaller proportion of tiles relative to background and healthy tissue, particularly at lower altitudes where individual leaves occupy a larger portion of the frame. Each model will be trained for up to 50 epochs, and the checkpoint with the lowest validation loss will be saved as the final model for that height. Final training epochs, batch size, per-height training time, and hardware configuration will be reported after all four models are trained.

3.5.1 Encoder: SegFormer MiT-B3 (MixVision Transformer)

The encoder component of the hybrid architecture employs the MixVisionTransformer-B3 (MiT-B3), originally introduced as part of the SegFormer family by Xie et al. (2021). MiT-B3 is a hierarchical Vision Transformer specifically designed for dense prediction tasks such as semantic segmentation, as opposed to standard Vision Transformers (ViT) which were originally designed for image classification.

The MiT-B3 encoder processes the 640 × 640 input image through four sequential stages, each producing feature maps at progressively reduced spatial resolutions (1/4, 1/8, 1/16, and 1/32 of the original input size). At each stage, three core operations are performed:

(a) Overlapping Patch Embedding: Unlike the original Vision Transformer, which divides the input into non-overlapping fixed-size patches, MiT-B3 uses overlapping patch embeddings implemented via strided convolutions. This overlap preserves local continuity between adjacent patches and prevents the boundary artifacts that non-overlapping tokenization introduces, which is critical for segmentation tasks where precise spatial boundaries between healthy and diseased tissue must be maintained.

(b) Efficient Self-Attention*:* Standard self-attention in Vision Transformers has a computational complexity that grows quadratically with the number of input tokens, making it prohibitively expensive for high-resolution images. MiT-B3 addresses this limitation through a spatial reduction mechanism that downsamples the key and value matrices before computing attention, reducing the computational cost while still capturing long-range global dependencies across the entire potato field. This global receptive field allows the encoder to consider the broader canopy context when classifying individual pixels, unlike convolutional networks that are limited to small local neighborhoods.

(c) Mix Feed-Forward Network (Mix-FFN): After the attention operation, each stage applies a Mix-FFN that incorporates a 3×3 depthwise convolution within the feed-forward layers. This convolution injects positional information directly into the feature representations, eliminating the need for explicit positional encoding that standard Transformers require. The depthwise convolution also reinforces local spatial relationships, complementing the global context captured by the self-attention mechanism.

The encoder is initialized with pre-trained ImageNet weights, providing a strong foundation of low-level visual features such as edges, textures, and color gradients. These pre-trained features are then fine-tuned on the domain-specific UAV potato field imagery during training. The four stages of the encoder output a set of hierarchical multi-scale feature maps that capture both fine-grained leaf-level textures (from the early, high-resolution stages) and broad field-level semantic context (from the deeper, low-resolution stages).

3.5.2 Decoder: U-Net

The decoder component follows the classical U-Net architecture, originally proposed by Ronneberger et al. (2015) for biomedical image segmentation. The primary function of the U-Net decoder is to reconstruct a full-resolution, pixel-wise segmentation map from the compressed, semantically rich feature representations produced by the MiT-B3 encoder.

The U-Net decoder operates through a series of progressive upsampling stages that mirror the encoder's downsampling stages in reverse. At each stage, two core operations are performed:

(a) Transposed Convolution (Upsampling*):* The spatially compressed feature maps from the encoder's deepest stage are progressively upsampled back toward the original 640 × 640 resolution. Each upsampling step doubles the spatial dimensions of the feature map while reducing the number of feature channels, gradually transitioning from abstract semantic representations to spatially precise predictions.

(b) Skip Connections: The defining characteristic of the U-Net architecture is the use of skip connections that concatenate the feature maps from the corresponding encoder stage with the upsampled decoder feature maps at each resolution level. This concatenation is critical because the encoder's downsampling process, while necessary for capturing high-level semantic information, inevitably discards fine spatial details such as the precise boundaries of disease lesions. The skip connections recover this lost spatial information by directly injecting the high-resolution encoder features into the decoder pathway. For potato disease segmentation, this mechanism is particularly important because disease lesions such as Early Blight and Late Blight often present as irregularly shaped regions with complex boundaries that require precise pixel-level delineation.

After the final upsampling stage, the decoder produces a feature map at the full 640 × 640 spatial resolution. A final 1×1 convolutional layer then projects this feature map into a four-channel output tensor of shape [Batch, 4, 640, 640], where each of the four channels contains the raw, unnormalized logit score for one of the four segmentation classes (Background, Healthy, Early Blight, and Late Blight) at every pixel location. During inference, an argmax operation is applied across the four class channels at each pixel to produce the final single-channel predicted segmentation mask.

The combination of the MiT-B3 encoder's global contextual understanding with the U-Net decoder's precise spatial reconstruction creates a hybrid architecture that leverages the complementary strengths of both Transformers and convolutional neural networks for accurate, pixel-level potato disease segmentation from UAV imagery.

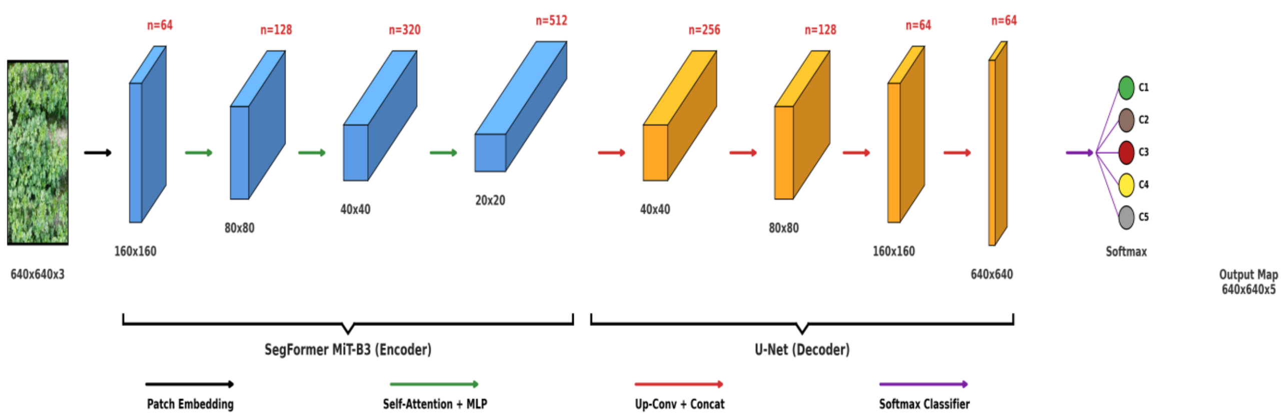

Fig 3.10: Abstract 3D representation of the Hybrid Transformer-CNN architecture.

3.6 Planned Model Evaluation

The four height-specific models will each be evaluated independently after training is completed, using their respective reserved test subsets. Because this is a semantic segmentation task, evaluation will be performed at the pixel level rather than at the image level. Per-height results will be reported individually so that performance differences attributable to image scale and GSD can be clearly identified. A cross-height comparison table will additionally summarise IoU, Dice score, F1-score, and disease detection rate across all four models. The evaluation design prioritises metrics that measure both disease detection accuracy and spatial boundary quality.

3.6.1 Confusion Matrix

A pixel-wise confusion matrix will be generated by comparing the predicted mask with the ground-truth mask. This will show how many pixels from each class are correctly or incorrectly predicted. The confusion matrix will be used to calculate accuracy, precision, recall, F1-score, and class-wise error patterns.

3.6.2 Accuracy

Accuracy will measure the proportion of correctly classified pixels across the test dataset. However, because UAV images may contain large amounts of background and healthy tissue, accuracy will not be treated as the only performance indicator.

Accuracy = (TP + TN) / (TP + TN + FP + FN) ----(1)

3.6.3 Precision

Precision will indicate how many pixels predicted as disease are actually disease pixels. High precision is important for reducing false alarms and unnecessary field intervention.

Precision = TP / (TP + FP) ----(2)

3.6.4 Recall

Recall will indicate how many true disease pixels are detected by the model. In UAV-based crop monitoring, recall is especially important because missed disease regions may allow infection to spread before intervention.

Recall = TP / (TP + FN) ----(3)

3.6.5 F1-Score

The F1-score will combine precision and recall into a single balanced measure. This is useful when disease pixels are less frequent than background or healthy tissue pixels.

F1-score = 2 x (Precision x Recall) / (Precision + Recall) ----(4)

3.6.6 Intersection over Union (IoU) and Mean IoU

Intersection over Union will measure the spatial overlap between the predicted disease region and the ground-truth mask. Mean IoU will be calculated by averaging IoU across all target classes. This metric is central to segmentation because it evaluates whether the predicted mask occupies the correct spatial region.

IoU = TP / (TP + FP + FN) ----(5)

3.6.7 Dice Similarity Coefficient

The Dice Similarity Coefficient will be used as an additional overlap metric. It is particularly useful for imbalanced segmentation problems where disease regions may occupy a small portion of the image.

Dice = (2 x TP) / (2 x TP + FP + FN) ----(6)

3.6.8 Cross-Height Model Comparison

After all four height-specific models are evaluated on their individual test subsets, a consolidated cross-height comparison will be produced. This comparison will present IoU, mean IoU, Dice score, precision, recall, and F1-score for each of the four models in a single table (see Table 3.6 in Section 3.7). The comparison will identify which altitude model achieves the highest overall segmentation performance, which class presents the most difficulty at each height, and whether disease detection accuracy degrades at higher altitudes where GSD is coarser and individual leaf detail is reduced. This per-height evaluation is a primary methodological contribution of the study because it provides quantitative evidence of how UAV operating height affects deep learning segmentation reliability in real agricultural field conditions.

3.7 Final Result Reporting Framework

After all four height-specific models are trained and evaluated, final results will be reported using structured tables so that per-height model performance, class-wise behaviour, and field-level disease severity can be clearly interpreted. The following tables define the reporting format to be completed after training and testing are finished for all four altitude models.

Table 3.5: Final Training Configuration

Parameter	Final Value
Model architecture	Hybrid Transformer-CNN (SegFormer MiT-B3 + U-Net)
Input tile size	640 x 640 pixels
Optimizer	AdamW
Learning rate	0.0001
Loss function	To be finalized after training setup
Batch size	[To be inserted]
Epochs	[To be inserted]
Training hardware	[To be inserted]

Training time	[To be inserted]

Table 3.6: Per-Height Segmentation Performance (Per-Model Test Results)

Metric	2m Model	7m Model	10m Model	12m Model
Overall Pixel Accuracy (%)	—	—	—	—
Mean IoU (%)	—	—	—	—
Background IoU (%)	—	—	—	—
Healthy IoU (%)	—	—	—	—
Early Blight IoU (%)	—	—	—	—
Late Blight IoU (%)	—	—	—	—
Healthy Precision (%)	—	—	—	—
Healthy Recall (%)	—	—	—	—
Healthy F1-Score (%)	—	—	—	—
Early Blight Precision (%)	—	—	—	—
Early Blight Recall (%)	—	—	—	—
Early Blight F1-Score (%)	—	—	—	—
Late Blight Precision (%)	—	—	—	—
Late Blight Recall (%)	—	—	—	—
Late Blight F1-Score (%)	—	—	—	—
Training Images	—	—	—	—
Test Images	—	—	—	—
Training Epochs	50	50	50	50

Table 3.7: Cross-Height Model Comparison Summary

Class	Precision	Recall	F1-score	IoU	Dice
Background	[After testing]	[After testing]	[After testing]	[After testing]	[After testing]
Healthy Tissue	[After testing]	[After testing]	[After testing]	[After testing]	[After testing]
Blight / Disease	[After testing]	[After testing]	[After testing]	[After testing]	[After testing]
Mean	[After testing]	[After testing]	[After testing]	[After testing]	[After testing]

Table 3.8: Disease Severity Output

Severity Level	Disease Density	Interpretation	Recommended Action
MINOR	< 10%	Trace or low-level infection	Routine monitoring
WARNING	10-30%	Emerging hotspot	Localized inspection or treatment
CRITICAL	> 30%	Severe disease concentration	Immediate intervention